

Decomposition, Smoothing, and Business Cycle Analysis

Chapter 2

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Outline

- Time Series Decomposition
- Moving Averages
- Linear filters
- Exponential Smoothing
- X-13-ARIMA-SEATS
- What's a Business Cycle?

Time Series Decomposition

Section 1

Time Series Decomposition

- Any series $\{y_t\}$ has three components or patterns
 - Trend (T) – the slow-moving trajectory
 - Seasonality (S) – a repeated calendar rhythm
 - Irregular (I) – what's left after trend and seasonality are removed
 - $I_t = y_t - T_t - S_t$

Time Series Decomposition

The additive model

- Assumes seasonality has a fixed **magnitude** regardless of the series level
 - a \$200B Q4 bump is the same whether GDP is \$10T or \$25T
- $y_t = T_t + S_t + I_t$

The multiplicative model

- Assumes seasonality scales with the level – 5% above/below trend
- $y_t = T_t \times S_t \times I_t$
- You can transform the multiplicative model into an additive one by taking logs
- $\log y_t = \log T_t + \log S_t + \log I_t$

Time Series Decomposition

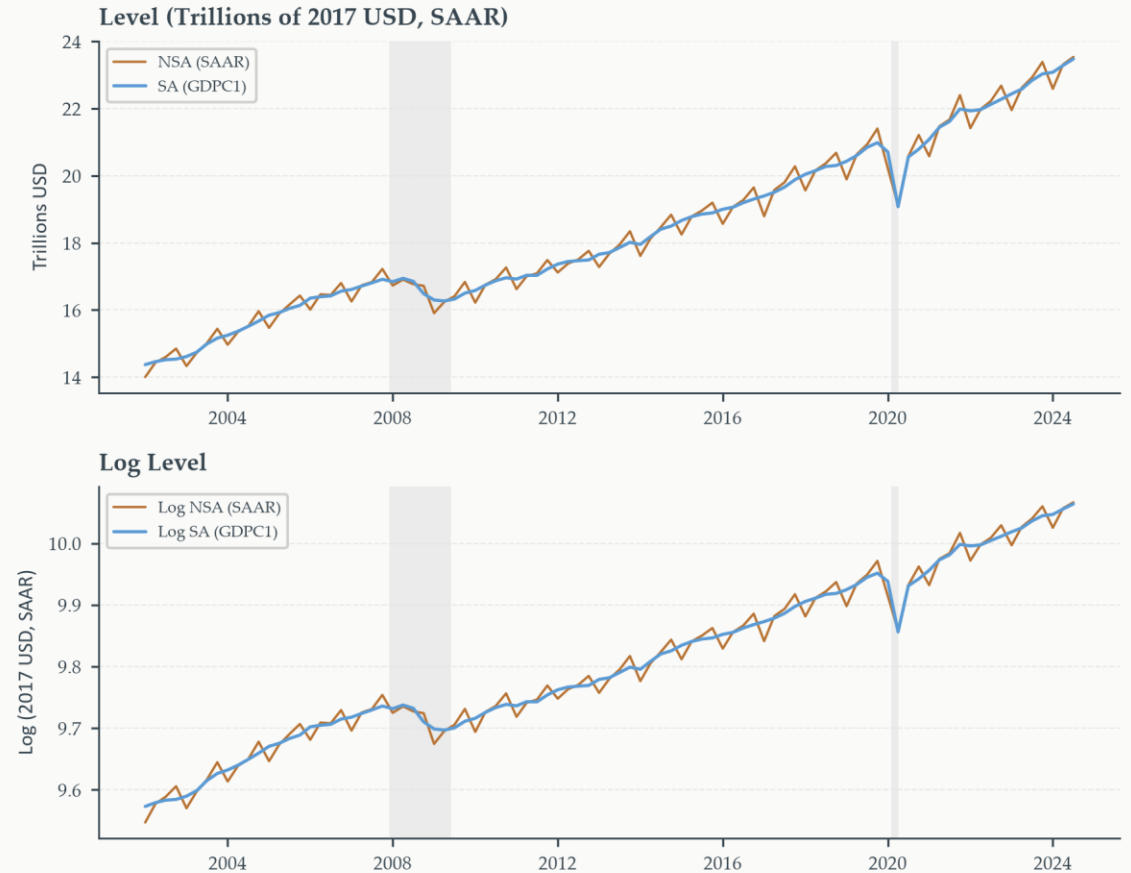
Top: levels

- Oscillations around the trend are of growing magnitude

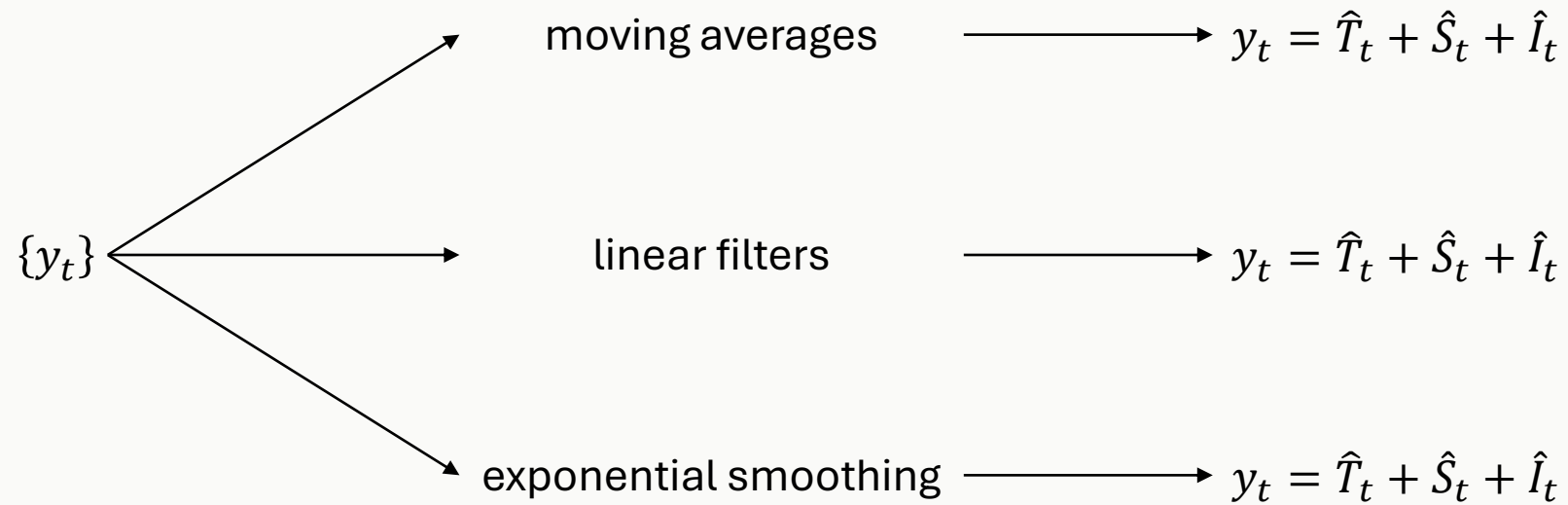
Bottom: log levels

- Oscillations are proportionally stable

US Real GDP: NSA vs. SA, 2002–2024



Time Series Decomposition



Moving Averages

Section 2

Moving Averages

How do we estimate a “slow-moving trajectory” from noisy quarterly data?

The oldest, most transparent approach: Replace each observation with a **local average** of the surrounding observations.

Short-run wiggles cancel; whatever survives the averaging is the trend

Whatever the moving average misses becomes the seasonal and irregular components – trend estimation and decomposition are the same operation

Moving Averages

- The 2x4 Moving Average

- Assume data with 4 quarters
- Build a moving average in **three steps**
- Step 1: Forward moving average

- $M_t^+ = \frac{1}{4}(y_{t-1} + y_t + y_{t+1} + y_{t+2})$

- Step 2: Backward moving average

- $M_t^- = \frac{1}{4}(y_{t-2} + y_{t-1} + y_t + y_{t+1})$

- Step 3: Final moving average

- $\hat{T}_t = \frac{1}{2}(M_t^+ + M_t^-) = \frac{1}{8}y_{t-2} + \frac{1}{4}y_{t-1} + \frac{1}{4}y_t + \frac{1}{4}y_{t+1} + \frac{1}{8}y_{t+2}$

Moving Averages

- The 2x4 Moving Average (cont...)

- $\hat{T}_t = \frac{1}{8}y_{t-2} + \frac{1}{4}y_{t-1} + \frac{1}{4}y_t + \frac{1}{4}y_{t+1} + \frac{1}{8}y_{t+2}$

- Note the following two features

- $\hat{T}_t$ is centered at t

- Same weight for each quarter

- $\hat{T}_t = \frac{1}{8}Q_2^t + \frac{1}{4}Q_3^t + \frac{1}{4}Q_4^t + \frac{1}{4}Q_1^{t+1} + \frac{1}{8}Q_2^{t+1}$

Moving Averages

- The 2x4 Moving Average (cont...)
 - Once you have $\hat{T}_t$, you can decompose $\{y_t\}$
 - Step 1: $(\hat{S}_t + \hat{I}_t) = y_t - \hat{T}_t$
 - Step 2: $\hat{S}_Q = \frac{1}{n} \sum_1^n S_Q$ — average all de-trended values for each quarter and normalize to sum to zero
 - Step 3: $\hat{I}_t = \hat{y}_t - \hat{T}_t - \hat{S}_t$

Moving Averages

Top

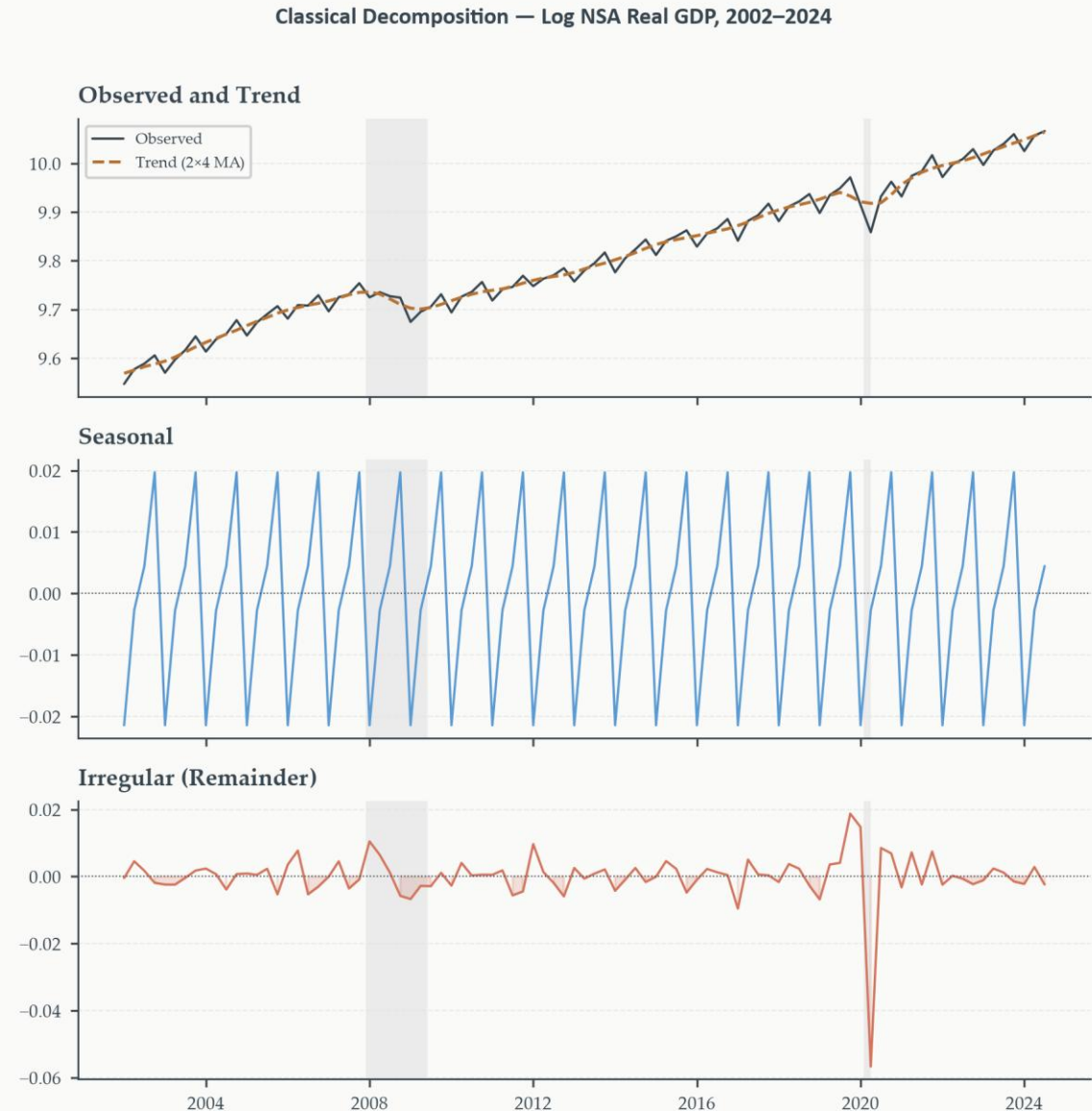
- Observed data and 2x4 MA trend

Middle:

- Seasonal factor; large and stable

Bottom:

- Irregular component, where the business cycle lives



Moving Averages

- The 2x4 Moving Average (cont...)
 - Limitations: There is no optimization
 - The number of periods (quarters) to average is fixed
 - The weights for each period (quarter) are fixed
 - There is no smooth adjustment to the trend

Linear filters

Section 3

Linear filters

- A simple baseline: Linear (and quadratic) detrending
 - $\log y_t = \alpha + \beta t + \varepsilon_t$
 - $\log y_t = \alpha + \beta t + \gamma t^2 + \varepsilon_t$
 - The residual (e) is the cyclical and irregular component.

Linear filters

- The **HP Filter** is the most widely used business-cycle tool in macroeconomics
 - Transparent optimization problem with a smooth trend

$$\min_{\{\tau_t\}} \left\{ \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2 \right\}$$

- **First sum:** penalizes deviation of trend from data — pulls the trend *toward* the data
- **Second sum:** penalizes changes in trend's *growth rate* — pulls the trend toward a straight line
- λ : governs the tradeoff

Linear filters

- Where does λ come from?
 - It is exogenous to the minimization problem
 - The value of lambda comes from **convention**
 - Quarterly data $\rightarrow \lambda = 1600$
 - Monthly data $\rightarrow \lambda = 14400$
 - Extreme values of λ
 - $\lambda \rightarrow 0$: $\{\tau_t\}$ passes through every datapoint (no smoothing)
 - $\lambda \rightarrow \infty$: $\{\tau_t\}$ approaches a straight line (maximum smoothing)
 - Filtering the series
 - $(\hat{S}_t + \hat{I}_t) = y_t - \tau_t$

Linear filters

- Hamilton's critique – *Why You Should **Never** Use the Hodrick-Prescott Filter*
 - **Spurious cycles.** Applying the HP filter to a *pure random walk* produces a cyclical residual with similar statistical properties to those reported from a real business cycle
 - **End-point bias.** The trend near the most recent observations is especially sensitive to λ ; precisely when policymakers need reliable estimates
 - **Wrong-order penalty:** The HP filter assumes that the data is $I(2)$, but log of GDP is $I(1)$ — right optimization problem for the wrong type of process

Linear filters

- Hamilton's critique (cont...)
 - Instead, run an OLS regression to forecast y_t h steps ahead; y_{t+h}
 - $y_{t+h} = \alpha + \beta_0 y_t + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 y_{t-3} + \varepsilon_{t+h}$
 - The residual ($e = y_{t+h} - \hat{y}_{t+h}$) is the cyclical (and irregular) component
 - Value of h ? Two years ahead (8 quarters)
 - Pros
 - No need to decide on a smoothing parameter (λ)
 - Doesn't fabricate cycles from noise
 - No end-point bias
 - Cons (?)
 - Assumes a fixed cycle frequency

Linear filters

- The End-Point Bias Problem

- Real-time

- Run data up to 2007 (just before the 2008 financial crisis); what happens?
 - The filter sees a long expansion with no future collapse to “know about”
 - The interpretation is that 2004-2007 is trend-growth, not a cycle about to self-correct

- Full sample

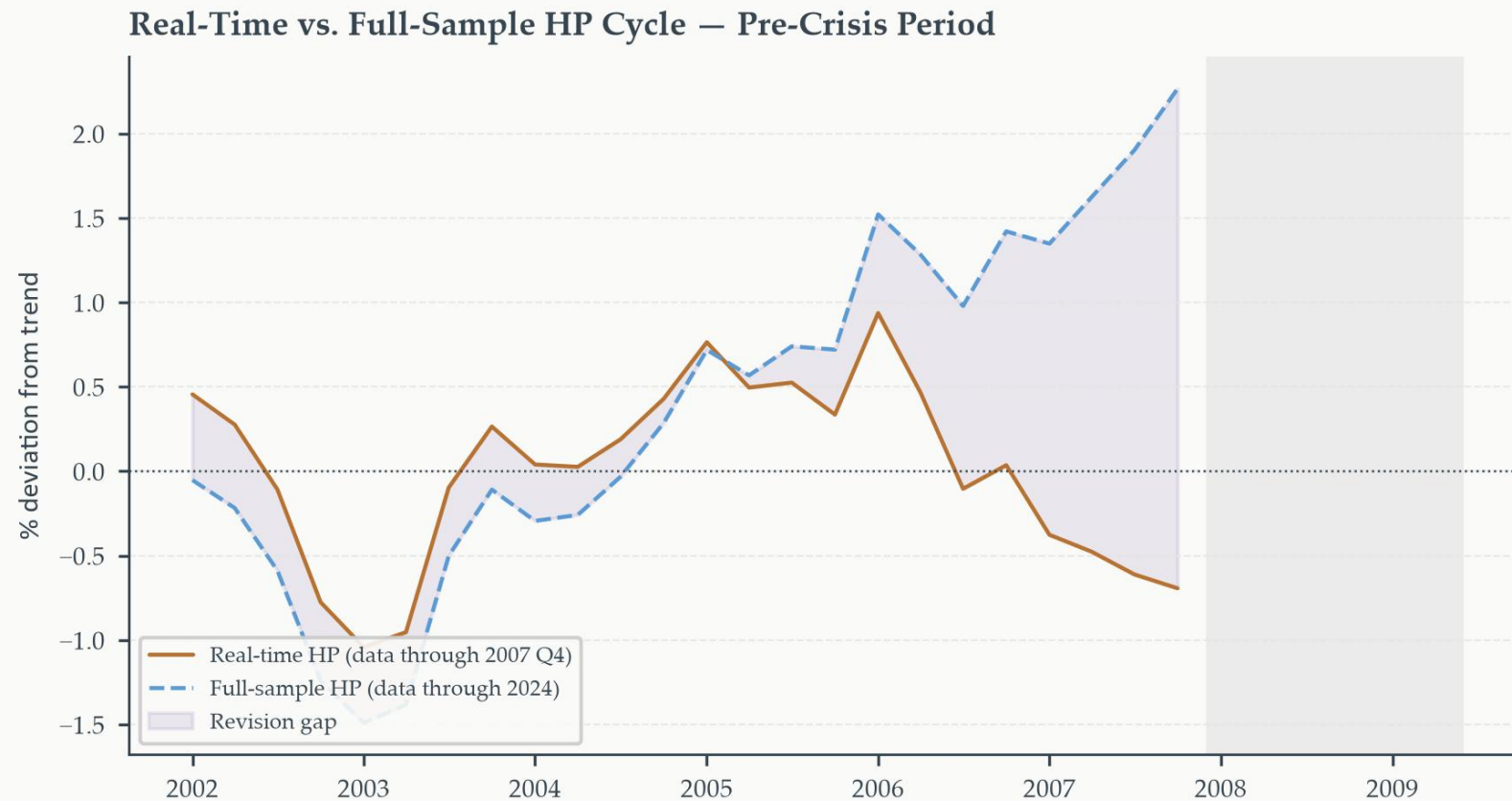
- Run data up to 2024 (include the 2008 financial crisis); what happens?
 - The Great Recession pulls the trend down, producing a larger cycle component

- Hamilton’s filter

- A cycle at date t depends only on observations well before t ; so that real-time and full-sample estimation **coincide** by construction

Linear filters

HP Filter End-Point Bias: Real-Time vs. Full-Sample, 1947–2007



Exponential Smoothing

Section 4

Exponential Smoothing

- Simple Exponential Smoothing (SES)
 - Update the estimate of today's value after each new observation
 - $\hat{y}_{t+1|t} = \hat{y}_{t|t-1} + \alpha e_t, \quad e_t = y_t - \hat{y}_{t|t-1} \wedge \alpha \in [0,1]$
 - $\hat{y}_{t+1|t} = \hat{y}_{t|t-1} + \alpha(y_t - \hat{y}_{t|t-1})$
 - Large α : aggressive error correction (updating)
 - Small α : conservative error correction
- Do some math-magic:
 - $\hat{y}_{t+1|t} = \alpha y_t + (1 - \alpha)\hat{y}_{t|t-1}$

Exponential Smoothing

- Simple Exponential Smoothing (SES) cont...)
 - Flat estimation: $\hat{y}_{T+h|T} = \hat{y}_{T+1|T} \quad \forall h \geq 1$
 - Appropriate for series with **no trend** and **no seasonality**

Exponential Smoothing

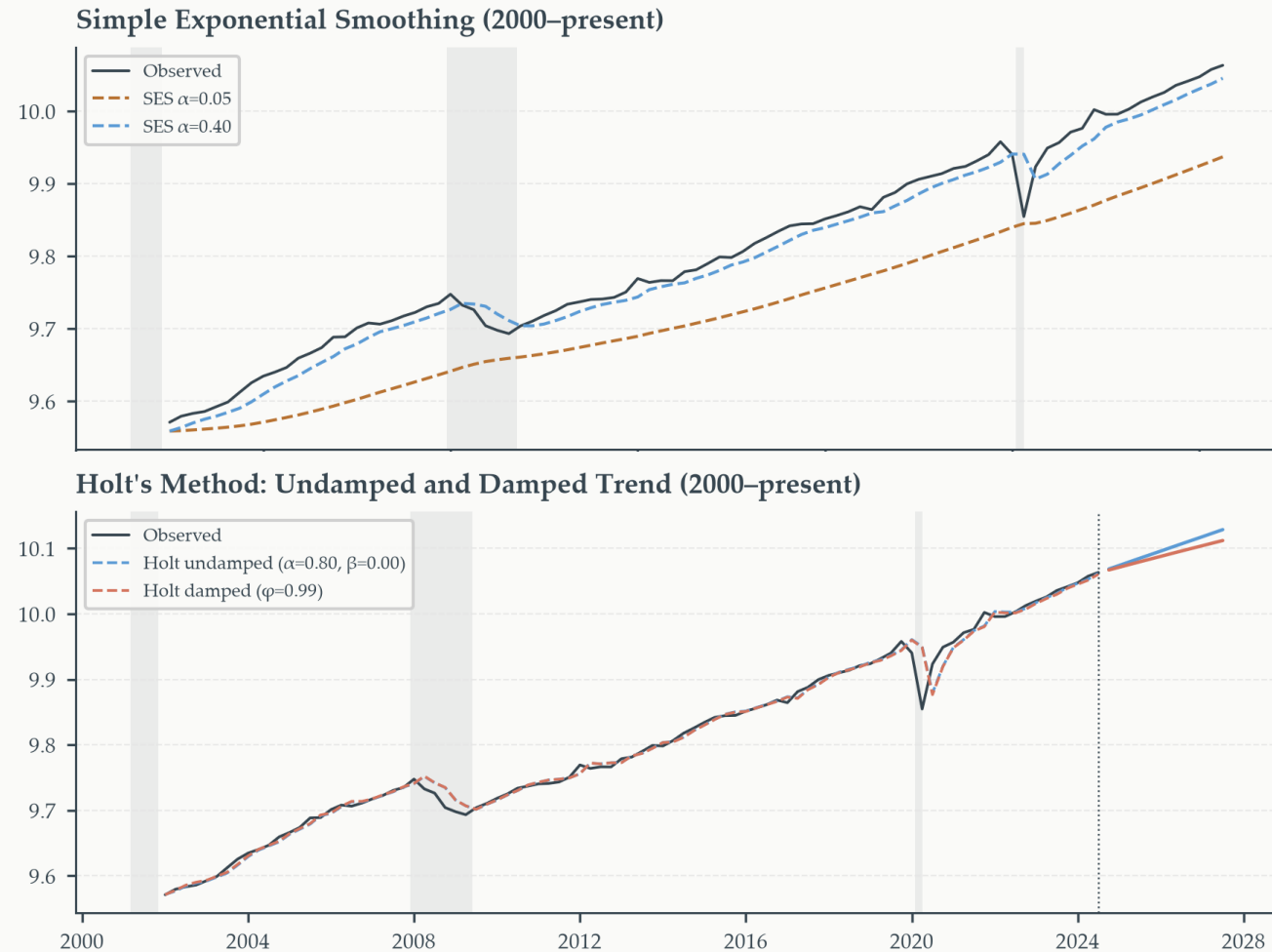
- Add trend: Holt's method
 - Let $\hat{\ell}_t$ be the estimated **level**
 - Let $\hat{b}_t$ be the estimated **change** in level between t and $t - 1$
 - Then:

$$\begin{aligned}\hat{y}_{t+h|t} &= \hat{\ell}_t + \hat{b}_t h \\ \hat{\ell}_t &= \alpha y_t + (1 - \alpha)(\hat{\ell}_{t-1} + \hat{b}_{t-1}) \\ \hat{b}_t &= \beta(\hat{\ell}_t - \hat{\ell}_{t-1}) + (1 - \beta)\hat{b}_{t-1}\end{aligned}$$

Exponential Smoothing

- Correct for absurd growth: Holt's dampened trend
 - Add a damping factor $\phi \in (0,1)$ at each horizon:
 - $\hat{y}_{t+h|t} = \hat{\ell}_t + (\phi + \phi^2 + \dots + \phi^h)\hat{b}_t$
 - As $h \rightarrow \infty$: $\sum \phi^i \rightarrow \frac{\phi}{1-\phi} < \infty$

Exponential Smoothing



Exponential Smoothing

- Add Seasonality: Holt-Winters
 - Let $\hat{s}_t$ be the smoothed **seasonal factor**

$$\begin{aligned}\hat{y}_{t+h|t} &= \hat{\ell}_t + \hat{b}_t h + \hat{s}_{t+h-m(k+1)} \\ \hat{\ell}_t &= \alpha(y_t - \hat{s}_{t-m}) + (1 - \alpha)(\hat{\ell}_{t-1} + \hat{b}_{t-1}) \\ \hat{b}_t &= \beta(\hat{\ell}_t - \hat{\ell}_{t-1}) + (1 - \beta)\hat{b}_{t-1} \\ \hat{s}_t &= \gamma(y_t - \hat{\ell}_{t-1} - \hat{b}_{t-1}) + (1 - \gamma)\hat{s}_{t-m}\end{aligned}$$

Exponential Smoothing

- A notation guide
 - Three series components
 - Four potential treatments
 - **Error:**
 - (A): additive | (M): multiplicative
 - **Trend:**
 - (N): none | (A): additive | (Ad): Additive damped | (M) multiplicative
 - **Seasonality**
 - (N): None | (A): additive | (M): Multiplicative

ETS(E,T,S) identifies a specific model

ETS(A,N,N) = SES | ETS(A,A,N) = Undamped Holt | etc...

X-13-ARIMA-SEATS

Section 5

X-13-ARIMA-SEATS

- Standard government method: how does it work?
 - X-11: Original software with an iterative moving-average filter
 - Estimate the trend through an iterative process
 - X-13-ARIMA-SEATS
 - Current software version

X-13-ARIMA-SEATS

- The X-13-ARIMA-SEATS process (alternative 1)
 - Step 1: ARIMA extension
 - Step 2: Correct for deterministic effects (holidays, outliers, ...)
 - Step 3: X-11 filter
 - Extract a trend and seasonality from “Step 2” through a moving average technique
 - Repeat (the iterative process) until estimates converge (two or three passes)
 - Step 4: Discard the ARIMA extensions

X-13-ARIMA-SEATS

- The X-13-ARIMA-SEATS process (alternative 2)
 - Step 1: ARIMA extension
 - Step 2: Correct for deterministic effects (holidays, outliers, ...)
 - Step 3: Apply a SEATS (Signal Extraction ARIMA Time Series) model
 - Model-based decomposition instead of X-11 iterative process
 - Step 4: Discard the ARIMA extensions

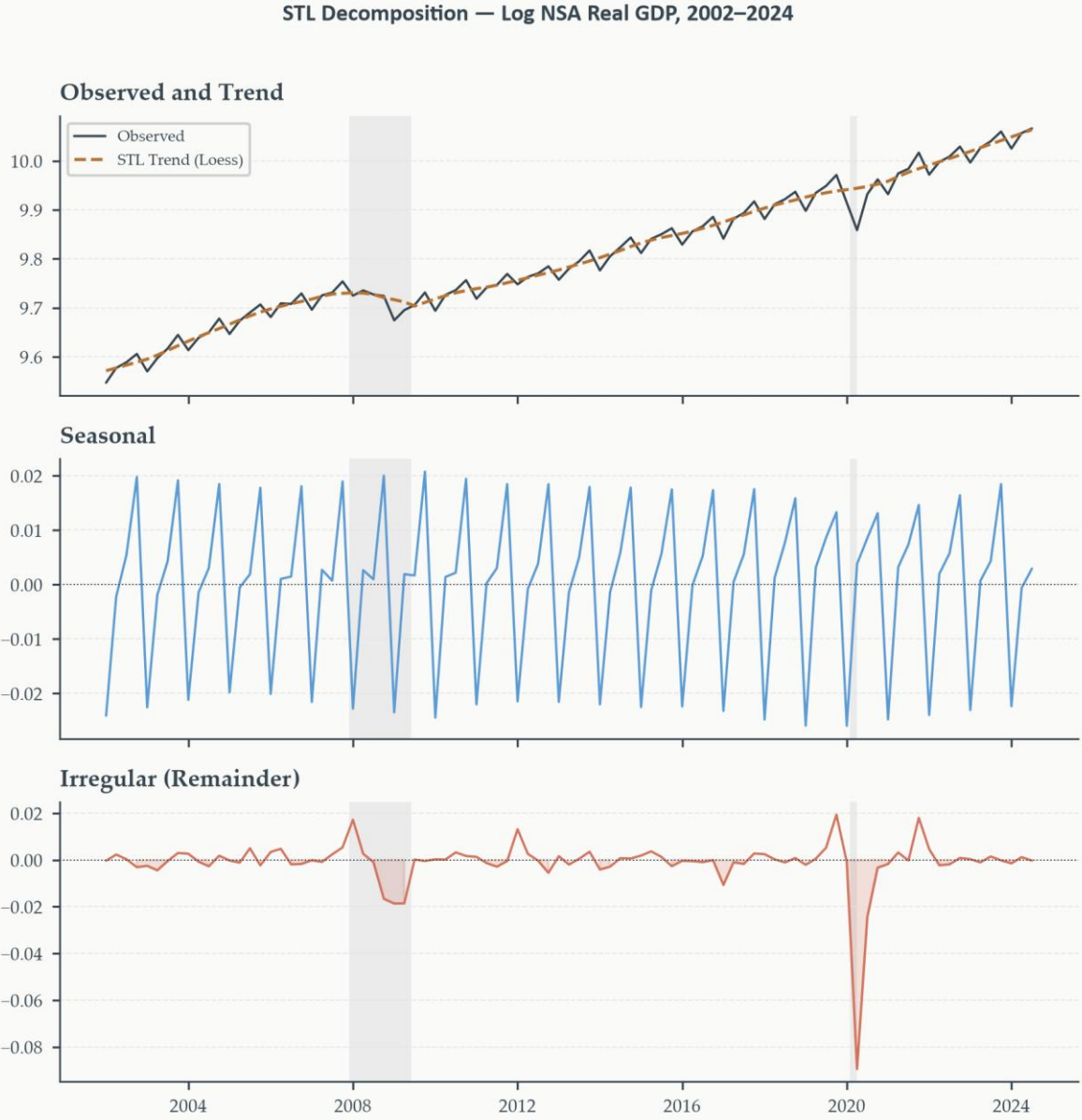
X-13-ARIMA-SEATS

- STL decomposition
 - X-13-ARIMA-SEATS is difficult to replicate due to its precise industrial grade and specific adjustment to deterministic factors
 - Alternative: Estimate a Seasonal-Trend Loess (STL) decomposition model

X-13-ARIMA-SEATS

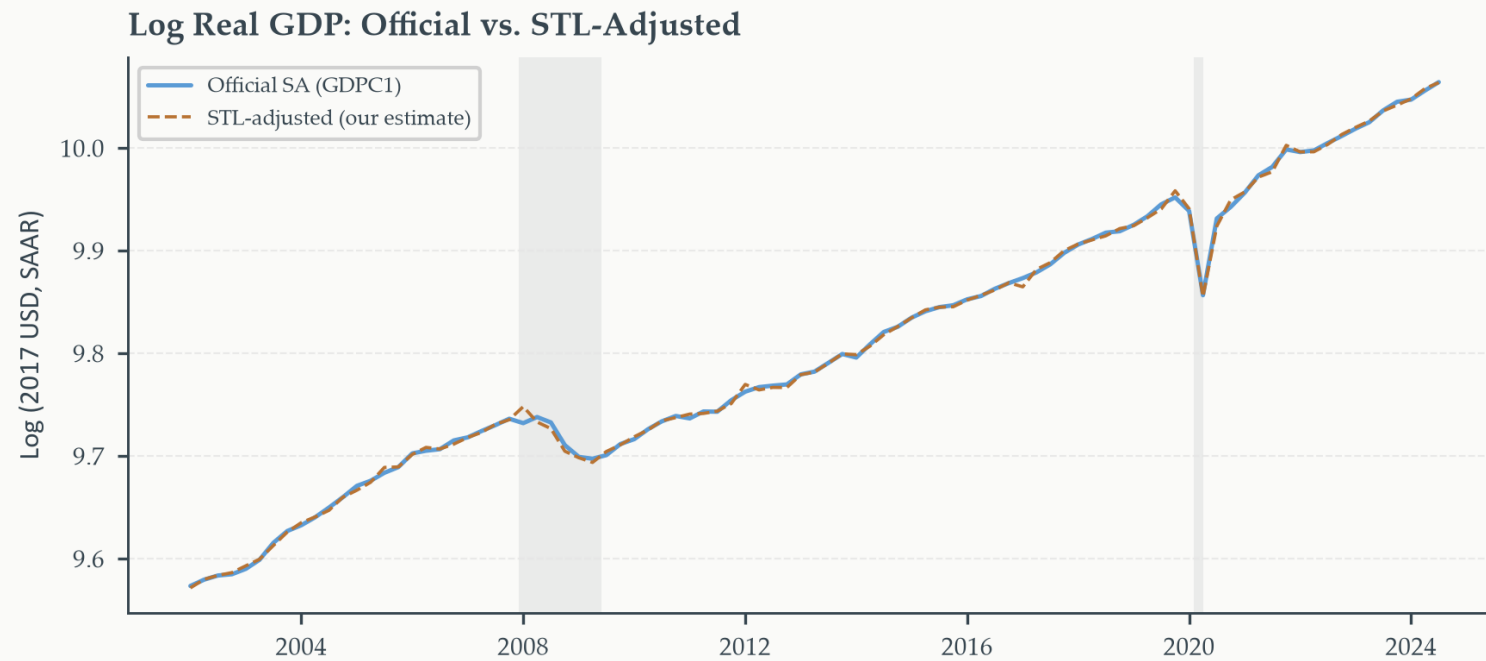
- STL decomposition (cont...)
 - Two iterative processes
 - Inner loop: Updates the seasonal and trend estimates
 - Outer loop: Computes robustness weights to reduce the influence of outliers
 - **Loess**: Estimate the trend by fitting a low-degree polynomial to a local neighborhood of the data at each point

X-13-ARIMA-SEATS



X-13-ARIMA-SEATS

STL Seasonal Adjustment vs. Official GDPC1, 2002–2024



STL closely tracks BEA's X-13-ARIMA-SEATS real GDP adjustment

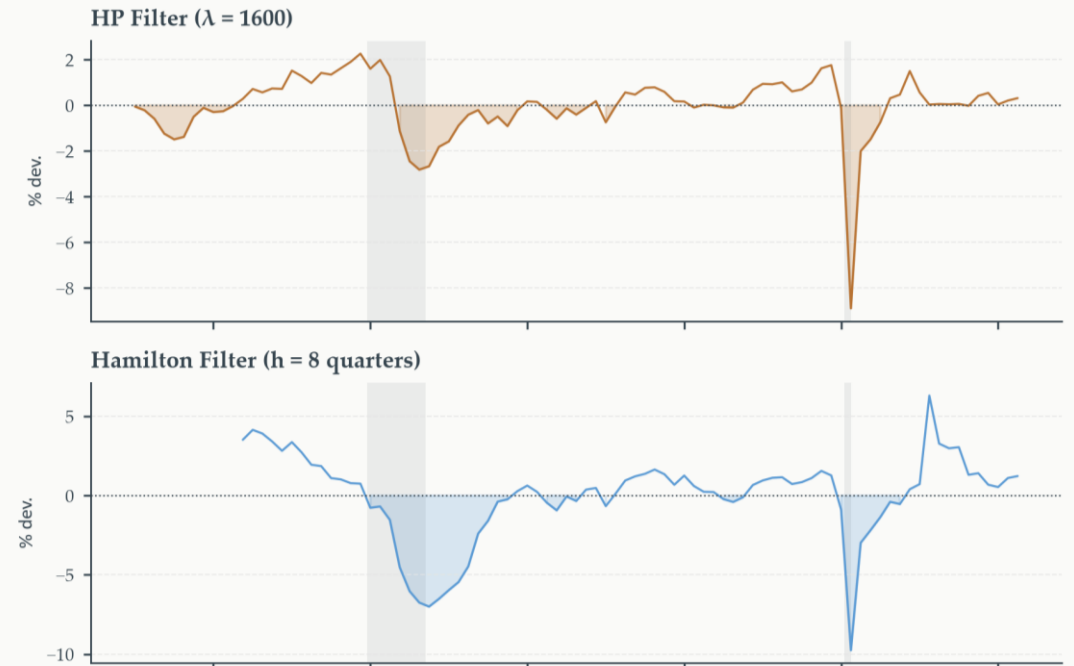
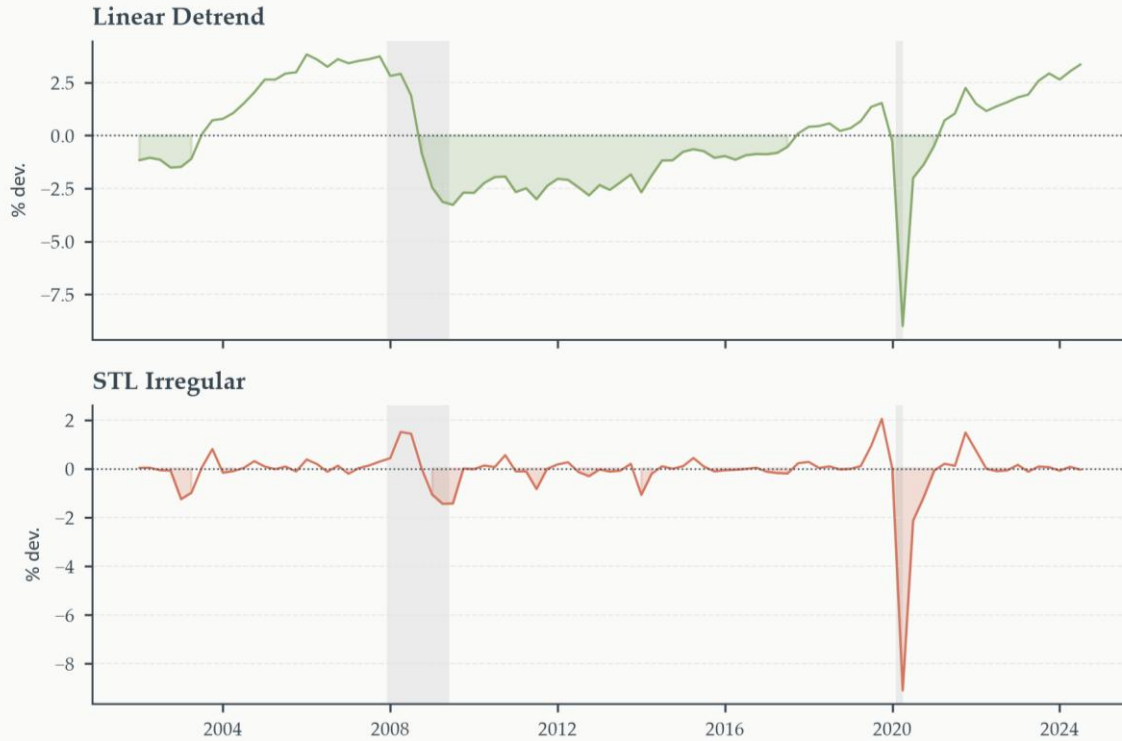
What's a Business Cycle?

Section 6

What's a business cycle?

- Deviations from the trend **net** of the seasonal component
 - $I_t = y_t - T_t - S_t$
 - The business cycle is not a “natural” object, is a statistical construction
 - And how this construction looks depends on the method used

What's a business cycle?



What's a business cycle?

- The Lucas' critique
 - Every detrending method embeds a structural assumption about potential output
 - HP filter: Trend is $I(2)$
 - Hamilton filter: Cycle is unpredictable two years ahead
 - Linear detrend: Growth rate is constant forever
 - When policymakers use a cycle estimate to calibrate a rule, they are conditioning policy on one of these assumptions
 - Lucas (1976): agents observe and respond to policy, altering the statistical relationships the model was built on
 - A trend estimate from pre-policy data may not be valid once the policy is known
 - **Bottom line:** The output gap is not just uncertain — it is endogenous to the policy that responds to it

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